

First Subject



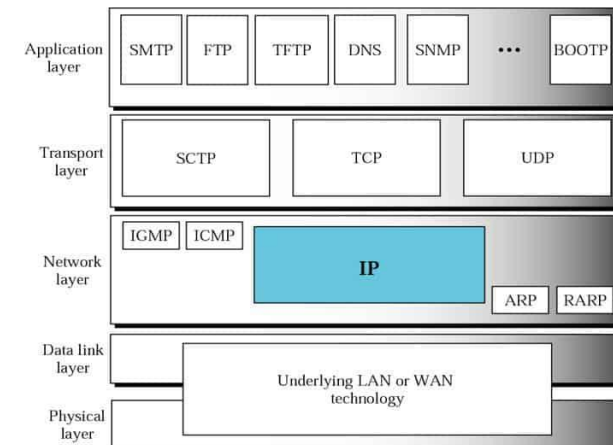
What is a Protocol?

- Agreed-upon communication rules
- Defines message format
- Specifies timing and order
- Like a language for computers

IP Internet Protocol

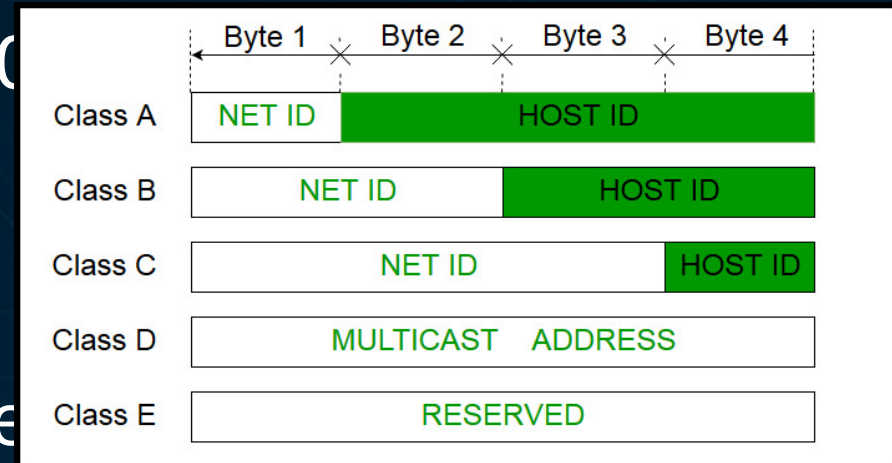
- IP (Internet Protocol)
- Layer 3 protocol
- Routes packets between networks
- Two versions: IPv4 and IPv6
- Example: 192.168.1.10

Position of IP in TCP/IP protocol suite



IPv4 Addressing

- IPv4 format: 192.168.1.10
- Four octets (0-255)
- 32 bits total
- About 4.3 billion addresses



- Classes: A, B, C (for different network sizes)

Special IP Addresses

- Private ranges (not routable):
 - 10.0.0.0 10.255.255.255
 - 172.16.0.0 172.31.255.255
 - 192.168.0.0 192.168.255.255
- Special:
 - 127.0.0.1 (localhost)
 - 0.0.0.0 (any address)



CYBERLABS

TCP Transmission Control Protocol

- TCP Reliable delivery:
 - Connection-oriented (handshake)
 - Guarantees delivery
 - Error checking
- Used for: Web, email, file transfer, SSH

```
03:06:31.546515 IP siftworkstation.ssh > 192.168.184.1.53548: Flags [P.], seq 41132:41504, ack 1801, win 761, options [nop,nop,TS val 18284
36746 ecr 2390120595], length 372
03:06:31.546991 IP 192.168.184.1.53548 > siftworkstation.ssh: Flags [.], ack 41504, win 2048, options [nop,nop,TS val 2390120698 ecr 182843
6746], length 0
03:06:31.651585 IP siftworkstation.ssh > 192.168.184.1.53548: Flags [P.], seq 41504:41876, ack 1801, win 761, options [nop,nop,TS val 18284
36851 ecr 2390120698], length 372
```


UDP User Datagram Protocol

- UDP Fast but unreliable:
 - Connectionless (no handshake)
 - No delivery guarantee
 - Faster than TCP
- Used for: Streaming, gaming, DNS, Service Discovery

```
03:11:38.796126 IP 192.168.184.1.57621 > 192.168.184.255.57621: UDP, length 44
03:11:54.273961 IP 192.168.184.1.63987 > 239.255.255.250.1900: UDP, length 171
03:11:55.274901 IP 192.168.184.1.63987 > 239.255.255.250.1900: UDP, length 171
03:11:56.276083 IP 192.168.184.1.63987 > 239.255.255.250.1900: UDP, length 171
03:11:57.276958 IP 192.168.184.1.63987 > 239.255.255.250.1900: UDP, length 171
03:12:08.135741 IP 192.168.184.1.60134 > 239.255.255.250.1900: UDP, length 125
```

Ports Subject



Port Numbers

- Ports identify applications:
 - 0-1023: Well-known ports
 - 1024-49151: Registered ports
 - 49152-65535: Dynamic/ephemeral

Think: Apartment building IP is street address,
port is apartment number. On top are hotel rooms.

Common Ports You Must Know

Port	Protocol	Service
20/21	FTP	File transfer service
22	SSH	Secure Shell
25	SMTP	Email sending (unencrypted!)
53	DNS	Domain Names
80	HTTP	Web (unencrypted!)
443	HTTPS	Web (encrypted)
3389	RDP	Windows Remote Desktop

ICMP Internet Control Message Protocol

- ICMP Network diagnostics:

- Layer 3 protocol
- No ports (not TCP/UDP)
- Error reporting

- Tools: ping, traceroute

```
ping nu.nl
```

```
PING nu.nl (118.69.16.182) 56(84) bytes of data.  
64 bytes from 118.69.16.182: icmp_seq=1 ttl=128 time=6.31 ms  
64 bytes from 118.69.16.182: icmp_seq=2 ttl=128 time=6.21 ms  
64 bytes from 118.69.16.182: icmp_seq=3 ttl=128 time=4.78 ms
```

```
03:06:30.091568 IP siftworkstation > 118.69.16.182: ICMP echo request, id 21023, seq 7, length 64  
03:06:30.096473 IP 118.69.16.182 > siftworkstation: ICMP echo reply, id 21023, seq 7, length 64
```

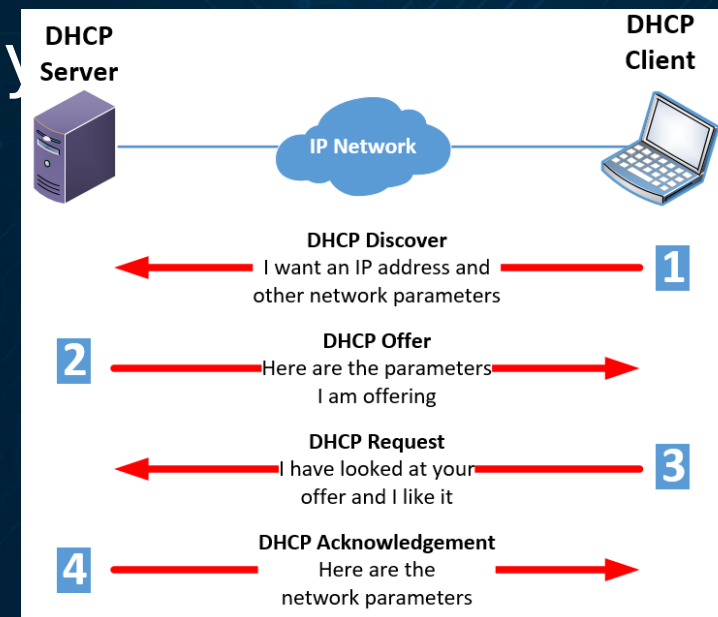
ARP Address Resolution Protocol

- ARP Maps IP to MAC:
 - Operates at Layer 2
 - "Who has this IP address?"
 - Builds ARP cache/table
- Security risk: ARP spoofing

```
03:05:32.714613 ARP, Request who-has _gateway tell siftworkstation, length 28
03:05:32.714860 ARP, Request who-has 192.168.184.254 tell siftworkstation, length 28
03:05:32.714963 ARP, Reply _gateway is-at 00:50:56:f1:6b:6e (oui Unknown), length 46
03:05:32.715170 ARP, Reply 192.168.184.254 is-at 00:50:56:ef:b3:d3 (oui Unknown), length 46
```

DHCP Dynamic Host Configuration Protocol

- DHCP Automatic IP assignment:
 - DORA process (Discover, Offer, Request, Acknowledge)
- Assigns IP, subnet, gateway
 - Temporary leases
 - Ports: UDP 67/68



DNS Domain Name System

- DNS Internet's phonebook:
 - Converts names to IPs
 - Hierarchical system
- Port: UDP 53 (TCP for large queries)
 - Example: google.com → 142.250.80.46

HTTP vs HTTPS

- HTTP HyperText Transfer Protocol:
 - Port 80
 - Unencrypted ⚠️
 - Plaintext visible
- HTTPS HTTP Secure:
 - Port 443
 - Encrypted with TLS/SSL ✅
 - Protects privacy

SSH Secure Shell

- SSH Secure remote access:
 - Port 22
 - Encrypted command line
 - Replaces insecure Telnet
 - Used for: Linux administration, file transfer

SMTP, POP3, IMAP Email Protocols

- Email protocols:
 - SMTP (25): Sending email
 - POP3 (110): Download & delete
 - IMAP (143): Sync across devices
- Secure versions: 465/587 (SMTPS), 995 (POP3S), 993 (IMAPS)

SNMP Simple Network Management Protocol

- SNMP Network device management:
 - Monitor routers, switches, servers
 - Ports: UDP 161/162
 - Versions: v1, v2c, v3
 - Security: Use SNMPv3 (encrypted)

Protocol Summary Security Perspective

- Secure (encrypted):
-  HTTPS, SSH, SFTP, IMAPS
- Insecure (plaintext):
-  HTTP, Telnet, FTP, SMTP
- Reconnaissance:
-  ICMP, DNS, SNMP

Network Tools



Essential Network Tools

- Windows & Linux tools we'll master:
 - ping (connectivity testing)
 - traceroute/tracert (path tracing)
 - ipconfig/ifconfig (configuration)
 - netstat (connection monitoring)
 - nslookup/dig (DNS queries)
 - Wireshark (traffic analysis)

ping Connectivity Testing

- ping Basic connectivity test:
 - Uses ICMP protocol
 - Measures round-trip time
 - Tests reachability
 - Syntax: `ping [target]`

```
beheerder@siftworkstation ~ (57.097s)
ping nu.nl
PING nu.nl (118.69.16.182) 56(84) bytes of data.
64 bytes from 118.69.16.182: icmp_seq=1 ttl=128 time=6.31 ms
64 bytes from 118.69.16.182: icmp_seq=2 ttl=128 time=6.21 ms
64 bytes from 118.69.16.182: icmp_seq=3 ttl=128 time=4.78 ms
```

ping Practical Examples

- Common usage:
- Windows: `ping google.com -n 4`
- Linux: `ping google.com -c 4`
- What to check:
 - Reply received? (connectivity)
 - Time in ms (latency)
 - Packets lost? (reliability)

traceroute/tracert Path Tracing

- Traces route to destination:
 - Shows each hop (router)
 - Measures time per hop
 - Identifies network bottlenecks

- Windows: tracert
- Linux: traceroute

```
beheerder@siftworkstation ~ (20.187s)
traceroute cyberlabs-usa.com

traceroute to cyberlabs-usa.com (185.230.63.186), 64 hops max
 1  192.168.184.2  0.286ms  0.300ms  0.265ms
 2  192.168.1.1    3.258ms  2.400ms  2.352ms
 3  113.22.0.124   4.941ms  4.614ms  10.217ms
 4  * * *
 5  118.70.1.124   5.673ms  4.394ms  4.436ms
 6  169.254.205.1  5.158ms  4.230ms  4.542ms
 7
```



ipconfig/ifconfig /ip Network Configuration

- View network settings:
 - Windows: ipconfig /all
 - Linux: ifconfig or ip addr

- Shows:

- IP address
- Subnet mask
- Default gateway
- MAC address
- DNS servers (sometimes)

```
beheerder@siftworkstation ~ (0.036s)
ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:4c:c4:23 brd ff:ff:ff:ff:ff:ff
    altname enp2s0
    inet 192.168.184.130/24 metric 100 brd 192.168.184.255 scope global dynamic ens160
        valid_lft 1657sec preferred_lft 1657sec
    inet6 fe80::20c:29ff:fe4c:c423/64 scope link
        valid_lft forever preferred_lft forever
```

netstat Active Connections

- Monitor network connections:
 - netstat -ano (Windows)
 - netstat -tulpn (Linux)

- Shows:
 - Active connections
 - Listening ports
 - Process IDs
 - Connection state

```
ben@ben@networkstation:~$ sudo netstat -tulpn
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:445              0.0.0.0:*               LISTEN      1705/smbd
tcp        0      0 0.0.0.0:139              0.0.0.0:*               LISTEN      1705/smbd
tcp        0      0 0.0.0.0:22               0.0.0.0:*               LISTEN      1/init
tcp        0      0 127.0.0.1:631            0.0.0.0:*               LISTEN      18437/cupsd
tcp        0      0 127.0.0.53:53            0.0.0.0:*               LISTEN      1021/systemd-resolv
tcp        0      0 127.0.0.54:53            0.0.0.0:*               LISTEN      1021/systemd-resolv
tcp6       0      0 :::445                   :::*                    LISTEN      1705/smbd
tcp6       0      0 :::139                   :::*                    LISTEN      1705/smbd
tcp6       0      0 :::22                    :::*                    LISTEN      1/init
```


nslookup/dig DNS Queries

- Query DNS servers:
 - Both OS: nslookup [domain]
 - Linux (better): dig [domain]
- Purpose:
 - Resolve domain to IP
 - Test DNS configuration
 - Investigate DNS issues
 - Security research

nslookup/dig DNS Queries

```
dig cyberlabs-usa.com
```

```
; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> cyberlabs-usa.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 43378
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;cyberlabs-usa.com.                IN      A

;; ANSWER SECTION:
cyberlabs-usa.com.      5       IN      A      185.230.63.171
cyberlabs-usa.com.      5       IN      A      185.230.63.186
cyberlabs-usa.com.      5       IN      A      185.230.63.107

;; Query time: 116 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sun Oct 26 03:26:33 UTC 2025
;; MSG SIZE rcvd: 94
```

arp ARP Table Management

- View/modify ARP cache:
 - arp -a (view all)
 - arp -d (delete entry)
- Shows: IP to MAC address mappings
- Security use: Detect ARP spoofing

```
03:05:32.714613 ARP, Request who-has _gateway tell siftworkstation, length 28
03:05:32.714860 ARP, Request who-has 192.168.184.254 tell siftworkstation, length 28
03:05:32.714963 ARP, Reply _gateway is-at 00:50:56:f1:6b:6e (oui Unknown), length 46
03:05:32.715170 ARP, Reply 192.168.184.254 is-at 00:50:56:ef:b3:d3 (oui Unknown), length 46
```

Wire Shark



Introduction to Wireshark

- Wireshark Packet analyzer:
 - Captures live network traffic
 - Analyzes packet contents
 - Decodes protocols automatically
- Essential SOC tool
- Free and open source

Why Wireshark Matters

- Critical for security:
 - Detect suspicious traffic
 - Analyze malware communication
 - Investigate incidents
 - Understand attack patterns
- Skill every SOC analyst needs

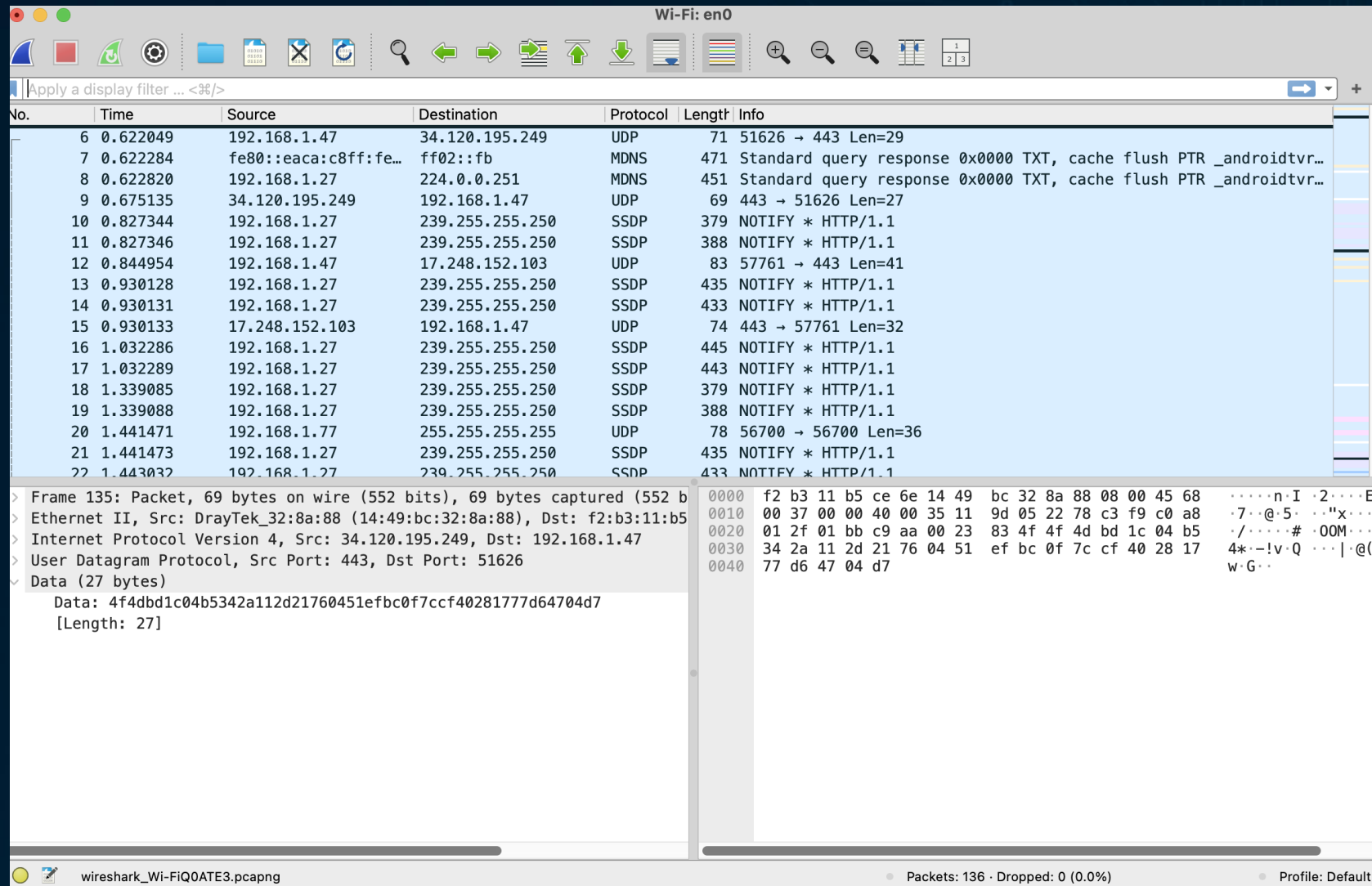
Wireshark Capabilities

- What Wireshark can do:
 - Capture live packets
 - Read PCAP files
 - Filter traffic (BPF)
 - Follow TCP streams
 - Export objects (files)
 - Statistics and graphs
 - Protocol dissection

Wireshark Interface Overview

- Main components:
 - Packet list (top)
 - Packet details (middle)
 - Packet bytes (bottom)
 - Filter bar (above packets)
 - Statistics menu

Wireshark Interface



Wi-Fi: en0

Apply a display filter ... <%%/>

No.	Time	Source	Destination	Protocol	Length	Info
6	0.622049	192.168.1.47	34.120.195.249	UDP	71	51626 → 443 Len=29
7	0.622284	fe80::eaca:c8ff:fe...	ff02::fb	MDNS	471	Standard query response 0x0000 TXT, cache flush PTR_androidtvr...
8	0.622820	192.168.1.27	224.0.0.251	MDNS	451	Standard query response 0x0000 TXT, cache flush PTR_androidtvr...
9	0.675135	34.120.195.249	192.168.1.47	UDP	69	443 → 51626 Len=27
10	0.827344	192.168.1.27	239.255.255.250	SSDP	379	NOTIFY * HTTP/1.1
11	0.827346	192.168.1.27	239.255.255.250	SSDP	388	NOTIFY * HTTP/1.1
12	0.844954	192.168.1.47	17.248.152.103	UDP	83	57761 → 443 Len=41
13	0.930128	192.168.1.27	239.255.255.250	SSDP	435	NOTIFY * HTTP/1.1
14	0.930131	192.168.1.27	239.255.255.250	SSDP	433	NOTIFY * HTTP/1.1
15	0.930133	17.248.152.103	192.168.1.47	UDP	74	443 → 57761 Len=32
16	1.032286	192.168.1.27	239.255.255.250	SSDP	445	NOTIFY * HTTP/1.1
17	1.032289	192.168.1.27	239.255.255.250	SSDP	443	NOTIFY * HTTP/1.1
18	1.339085	192.168.1.27	239.255.255.250	SSDP	379	NOTIFY * HTTP/1.1
19	1.339088	192.168.1.27	239.255.255.250	SSDP	388	NOTIFY * HTTP/1.1
20	1.441471	192.168.1.77	255.255.255.255	UDP	78	56700 → 56700 Len=36
21	1.441473	192.168.1.27	239.255.255.250	SSDP	435	NOTIFY * HTTP/1.1
22	1.443032	192.168.1.27	239.255.255.250	SSDP	433	NOTIFY * HTTP/1.1

Frame 135: Packet, 69 bytes on wire (552 bits), 69 bytes captured (552 b)

Ethernet II, Src: DrayTek_32:8a:88 (14:49:bc:32:8a:88), Dst: f2:b3:11:b5

Internet Protocol Version 4, Src: 34.120.195.249, Dst: 192.168.1.47

User Datagram Protocol, Src Port: 443, Dst Port: 51626

Data (27 bytes)

Data: 4f4dbd1c04b5342a112d21760451efbc0f7ccf40281777d64704d7 [Length: 27]

0000 f2 b3 11 b5 ce 6e 14 49 bc 32 8a 88 08 00 45 68n.I.2....Eh

0010 00 37 00 00 40 00 35 11 9d 05 22 78 c3 f9 c0 a8 .7..@.5.. "x....

0020 01 2f 01 bb c9 aa 00 23 83 4f 4f 4d bd 1c 04 b5 ./.....# .00M....

0030 34 2a 11 2d 21 76 04 51 ef bc 0f 7c cf 40 28 17 4*-!v.Q ...|@(.w.G..

0040 77 d6 47 04 d7

wireshark_Wi-FiQ0ATE3.pcapng

Packets: 136 · Dropped: 0 (0.0%)

Profile: Default

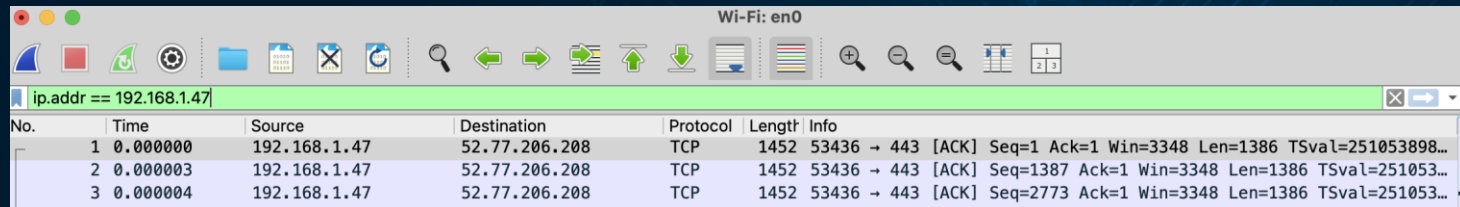
Capture Filters vs Display Filters

- Capture Filters:
 - Before capture starts
 - Less packets saved
 - BPF syntax
 - Example: ``port 80``
- Display Filters:
 - After capture
 - Filter what you see
 - Wireshark syntax
 - Example: ``http.request``

Common Display Filters

- Must know filters:

- ip.addr == 192.168.1.1
- tcp.port == 80
- http
- dns
- icmp
- tcp.flags.syn == 1



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.47	52.77.206.208	TCP	1452	53436 → 443 [ACK] Seq=1 Ack=1 Win=3348 Len=1386 TSval=251053898...
2	0.000003	192.168.1.47	52.77.206.208	TCP	1452	53436 → 443 [ACK] Seq=1387 Ack=1 Win=3348 Len=1386 TSval=251053...
3	0.000004	192.168.1.47	52.77.206.208	TCP	1452	53436 → 443 [ACK] Seq=2773 Ack=1 Win=3348 Len=1386 TSval=251053...

- Operators: == != && || !

Common Display Filters

Capturing from Wi-Fi: en0

http

No.	Time	Source	Destination	Protocol	Length	Info
2042	23.257446	192.168.1.47	192.168.1.27	HTTP	244	GET /ssdp/device-desc.xml HTTP/1.1
2045	23.262416	192.168.1.47	192.168.1.77	HTTP	224	GET / HTTP/1.1
2049	23.266776	192.168.1.77	192.168.1.47	HTTP/...	1275	HTTP/1.1 200 OK
2052	23.268620	192.168.1.27	192.168.1.47	HTTP/...	1252	HTTP/1.1 200 OK
2081	23.324428	192.168.1.47	192.168.1.77	HTTP	254	GET /apps/com.spotify.Spotify.TVv2 HTTP/1.1
2089	23.332669	192.168.1.47	192.168.1.27	HTTP	253	GET /apps/com.spotify.Spotify.TVv2 HTTP/1.1
2091	23.341321	192.168.1.27	192.168.1.47	HTTP	110	HTTP/1.1 404 Not Found
2103	23.362198	192.168.1.77	192.168.1.47	HTTP	217	HTTP/1.1 404 Not Found
2413	28.762562	192.168.1.47	1.1.1.1	HTTP	442	GET / HTTP/1.1
2416	28.786615	1.1.1.1	192.168.1.47	HTTP	452	HTTP/1.1 301 Moved Permanently (text/html)

> Frame 2045: Packet, 224 bytes on wire (1792 bits), 224 bytes captured (1792 bits) on interface en0
 > Ethernet II, Src: f2:b3:11:b5:ce:6e (f2:b3:11:b5:ce:6e), Dst: Arcadyan-802:11:11:11:11:11
 > Internet Protocol Version 4, Src: 192.168.1.47, Dst: 192.168.1.77
 > Transmission Control Protocol, Src Port: 56795, Dst Port: 1279, Seq: 1, Win: 0, Len: 0
 > Hypertext Transfer Protocol
 > GET / HTTP/1.1\r\n
 > Connection: keep-alive\r\n
 > Accept-Encoding: gzip\r\n
 > Keep-Alive: 0\r\n
 > Host: 192.168.1.77:1279\r\n
 > User-Agent: Spotify/127400477 OSX_ARM64/0 (Mac16,7)\r\n\r\n
 [Response in frame: 2049]
 [Full request URI: http://192.168.1.77:1279/]

```

0000 c0 d7 aa 8a 5b 18 f2 b3 11 b5 ce 6e 08 00 45 02  ....[... ..n..E..
0010 00 d2 00 00 40 00 40 06 b6 57 c0 a8 01 2f c0 a8  ..@.@..W.../..
0020 01 4d dd db 04 ff 31 4b be 3b a1 16 a3 de 80 18  .M...IK ;.....
0030 08 0b ba 68 00 00 01 01 08 0a 88 c8 b9 0d 00 08  .h.....
0040 a1 b0 47 45 54 20 2f 20 48 54 54 50 2f 31 2e 31  ..GET / HTTP/1.1
0050 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65  ..Connec tion: ke
0060 65 70 2d 61 6c 69 76 65 0d 0a 41 63 63 65 70 74  ep-alive ..Accept
0070 2d 45 6e 63 6f 64 69 6e 67 3a 20 67 7a 69 70 0d  -Encodin g: gzip
0080 0a 4b 65 65 70 2d 41 6c 69 76 65 3a 20 30 0d 0a  .Keep-Al ive: 0..
0090 48 6f 73 74 3a 20 31 39 32 2e 31 36 38 2e 31 2e  Host: 19 2.168.1.
00a0 37 37 3a 31 32 37 39 0d 0a 55 73 65 72 2d 41 67  77:1279 .User-Ag
00b0 65 6e 74 3a 20 53 70 6f 74 69 66 79 2f 31 32 37  ent: Spo tify/127
00c0 34 30 30 34 37 37 20 4f 53 58 5f 41 52 4d 36 34  400477 0 SX_ARM64
00d0 2f 30 20 28 4d 61 63 31 36 2c 37 29 0d 0a 0d 0a  /0 (Mac1 6,7)....
  
```


Following TCP Streams

- Reconstruct conversations:
 - Right-click packet
 - "Follow" → "TCP Stream"
 - See entire conversation
 - Use for: HTTP, email, FTP
- Color coded: Client vs Server

Following TCP Streams

No.	Time	Source	Destination	Protocol	Length	Info
7615	117.357989	192.168.1.4	142.250.199.195	QUIC	73	Protected Payload (K...
7616	117.370003	GrandstreamN_8b:61...	Broadcast	Ether...	564	Ethernet II
7617	117.381013	142.250.199.195	192.168.1.47	QUIC	65	Protected Payload (K...
7618	117.421321	192.168.1.47	142.250.71.202	UDP	71	55737 → 443 Len=29
7619	117.451500	142.250.71.202	192.168.1.47	UDP	67	443 → 55737 Len=25
7620	117.452553	192.168.1.47	192.168.1.27	TLSv1...	176	Application Data
7621	117.461025	192.168.1.27	192.168.1.47	TLSv1...	180	Application Data
7622	117.461026	192.168.1.27			66	8009 → 63425 [ACK] Seq=...
7623	117.461027	192.168.1.27			176	Application Data
7624	117.461110	192.168.1.47			66	63425 → 8009 [ACK] Seq=...
7625	117.461137	192.168.1.47			66	63425 → 8009 [ACK] Seq=...
7626	117.461412	192.168.1.47			176	Application Data
7627	117.466404	192.168.1.47			71	51620 → 443 Len=29
7628	117.466424	192.168.1.47			71	58537 → 443 Len=29
7629	117.499590	142.250.198.238			68	443 → 51620 Len=26
7630	117.499592	142.250.76.14			68	443 → 58537 Len=26
7631	117.509813	192.168.1.27			66	8009 → 63425 [ACK] Seq=...
7632	117.534046	192.168.1.47			71	59751 → 443 Len=29
7633	117.568527	142.250.198.234			67	443 → 59751 Len=25
7634	117.586310	208.103.161.2			603	Protected Payload (K...
7635	117.586313	208.103.161.2			603	Protected Payload (K...
7636	117.587075	192.168.1.47			85	Protected Payload (K...

Wireshark · Follow TCP Stream (tcp.stream eq 35) · Wi-Fi: en0

```

k...q.8.>.E.{..N.VR.@...i.j>l...|.i.f.C..k$h....x..G.d.Y.{S.IYg..S.-.\.k.
.m.9.w. Ah,M...)[..|k0. ..#hpV.....3.....%...=+.W.Y.(+...zL.\cIxen..z
x...H...F.LS...%0.d.....H..H.}> .....(.....H.}.9.....P.=...l....
D+[..p=...<...(.=.9
...l...rSS(..8...!C..N7.y.....j.....pzs5D..
g..1.|W.~;,...2u...C...e...r.H... (I.HSC....o.....(u|..\"D\
...m.V..za..Nq.H.<..n6..K...V..n.....\".4>...r.W.....7....1....D.*t.h.$...
}....M....5...:\Vf....=0..0...1m.....Q.d.....5.....^...5U....D.i.....!g.
-f..... G... KN....#B[.pi..Sq.0k...s.|Ts...2...p<..&7/.,e..@y.d\2a.l..
.#:....D...#.X... .#.Zr... .L.= { ...o. y...l.....r...v.:d.6t[.-...
7ur..FI.*.i.....<..~K.?L..pIF.....!.....g...D.8.....DM|; ...|I.#..
....I.....l_@.]G..`k.<.y
....I.<w....2z6.....a...{.....~0.@.L...-l.... =>Z..u..0...;5S..J.I{5...X.y
l#-0.u..k.....w.^..(.....'.S.....u...e...7=F.3G...<.Q.u..5.LZ.....9h.l....2.*
.J..A...4.....G..<...5.
bhpv...g.....p.....GP...h..d.ny*.V:.w..]B.S.g..+[[...\\8.....G(rS.C-..#...o8
.uE.0.0._4....q.Y.5C...f. .M.gE,?.@..B...rU.2...mED.y.}...#G'...|...T..
...U...o9y...t..q.p.0...5..la...V6...7...9...5.A;.....b...&. |E]>.*.vmQ...\"o.H.-v|
.....?5.W.vb...8.3pv>'+t....P.r. k..~....J...>.z$R.V.z.i.l_.....-..Sp...#..v-
...>^q.{6.7..o
.....C.%K&(7;.0.....'.%|.W'0...;z.....Z.....X.D.....Z]a8..
8...'P.....g.o1mu....\ o.K..C.S,.S1;.\].m..4F.0...Z...A...C.
...
.Z..Ez...F....6./r... (n.(.mD....^....#3.y.04..F..]0p...

d..F.#...)....
.EY.)..d..y....qn..>..V].D>z.,.....m]...6c...i..0..Fo@...F....p
...C0...kt.Aiu5.....pcR.....x.a.U.D.|[T. ....^3' p.cX:1..V.T...{.G.L
?..L9.

```

Packet 6329. 19 client pkts, 7 server pkts, 13 turns. Click to select.

Entire conversation (24 kB) Show as ASCII No delta times Stream 35

Find: Case sensitive Find Next

Help Filter Out This Stream Print Save as... Back Close

Exporting Objects from Capture

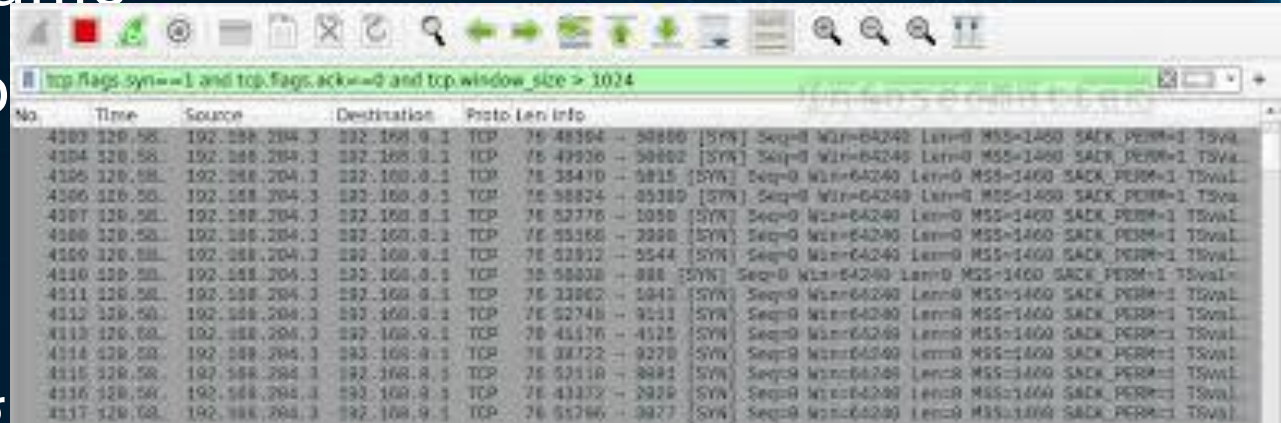
- Extract files from traffic:
 - File → Export Objects
 - Choose protocol (HTTP, SMB, etc.)
 - Save files transmitted
- Security use: Extract malware samples

Protocol Hierarchy Statistics

- Analyze traffic composition:
 - Statistics → Protocol Hierarchy
 - Shows all protocols captured
 - Percentage of traffic
 - Identify unusual protocols

Wireshark for Security Analysis

- Detect threats:
 - Malware C2 traffic
 - Data exfiltration
 - Port scans
 - ARP spoofing
 - DNS tunneling
 - Suspicious connections



tcp.flags.syn==1 and tcp.flags.ack==0 and tcp.window_size > 1024

No.	Time	Source	Destination	Proto	Len	Info
4303	320.58.	192.168.204.3	192.168.0.1	TCP	76	48364 → 50880 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4304	320.58.	192.168.204.3	192.168.0.1	TCP	76	49936 → 50880 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4305	320.58.	192.168.204.3	192.168.0.1	TCP	76	38470 → 50880 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4306	320.58.	192.168.204.3	192.168.0.1	TCP	76	38824 → 50880 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4307	320.58.	192.168.204.3	192.168.0.1	TCP	76	52776 → 1050 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4308	320.58.	192.168.204.3	192.168.0.1	TCP	76	53160 → 2080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4309	320.58.	192.168.204.3	192.168.0.1	TCP	76	52812 → 5544 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4310	320.58.	192.168.204.3	192.168.0.1	TCP	76	50038 → 888 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4311	320.58.	192.168.204.3	192.168.0.1	TCP	76	33862 → 5040 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4312	320.58.	192.168.204.3	192.168.0.1	TCP	76	52748 → 9211 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4313	320.58.	192.168.204.3	192.168.0.1	TCP	76	45176 → 4525 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4314	320.58.	192.168.204.3	192.168.0.1	TCP	76	38722 → 9270 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4315	320.58.	192.168.204.3	192.168.0.1	TCP	76	53110 → 8042 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4316	320.58.	192.168.204.3	192.168.0.1	TCP	76	43132 → 2920 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
4317	320.58.	192.168.204.3	192.168.0.1	TCP	76	53786 → 9027 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=

Wireshark Best Practices

- Use display filters early
- Save captures (PCAP format)
- Document findings
- Don't capture passwords (ethical!)
- Name saves descriptively
- Practice on safe networks first

Module 2 Theory What We Learned

- Network Models
 - OSI (7 layers)
 - TCP/IP (4 layers)
 - Encapsulation
- Protocols
 - IP, TCP, UDP
 - HTTP, DNS, SSH
 - Ports and services
- Tools
 - Basic: ping, tracer, ipconfig
 - Advanced: Wireshark

Key Takeaways

- Networks operate in layers
- Each protocol has a purpose
- Ports identify applications
- Tools help us analyze traffic
- Wireshark = security analyst's best friend

Security Mindset

- Think like an analyst:
 - What's normal traffic?
 - What looks suspicious?
 - Which protocols are risky?
 - Where can attacks hide?
- Network visibility = security

Common Protocol Ports

Quick Reference

Port	Protocol	Service
20/21	FTP	File transfer service
22	SSH	Secure Shell
25	SMTP	Email sending (unencrypted!)
53	DNS	Domain Names
80	HTTP	Web (unencrypted!)
443	HTTPS	Web (encrypted)
3389	RDP	Windows Remote Desktop

Next Up Hands-On Labs!

- 22 hours of practical work:
 - Network commands on Windows/Linux
 - Wireshark packet capture
 - Protocol analysis
 - Traffic filtering
 - Security investigations
 - Real-world scenarios

Questions to Consider

- Why do we need different layers?
- When would you use TCP vs UDP?
- What can Wireshark NOT do?
- How do filters help analysis?
- Why are plaintext protocols risky?

Thank you